



Requirements for this paper:

Multiple-choice
answer sheets:

☐

Non-programmable calculator:

☐

Graph paper:

☐

Laptop:

☒

Open book examination?

☒

EXAMINATION:	SEMESTER TEST ONE	PROGRAMME:	ADVANCED DIPLOMA ICT
MODULE CODE:	NDBS741	DURATION:	4 Hours
MODULE DESCRIPTION:	DATABASE SYSTEMS	MARKS:	50
EXAMINER(S):	DR SILAS VERKIJKA	DATE:	20/05/2024
MODERATOR(S):		TIME:	12:30

IMPORTANT INSTRUCTIONS

1. Answer ALL questions.
2. This is an open book Test. You may use any books or notes.
3. Laptop computers, tablets, and similar devices may be used.
4. You are allowed to use the internet during the Test.
5. You are not allowed to use a mobile phone or email application during the exam.
6. You are not allowed to use generative AI during the Test
7. You are not allowed to seek help or offer help to anyone during the Test.
8. Any attempt at cheating will be penalized.
9. You are allowed to ask question to the Lecturer.
10. Appendices will be provided for Test Data.

QUESTION 1 [7 MARKS]

You have been assigned the task of designing a Cassandra database to store information about various entities categorized by a "category" field. Each entity is uniquely identified by a combination of "item_id" and "category". The table should efficiently store and retrieve data using various data types, as described below:

- **item_id:** Unique identifier for an item.
- **category:** Category associated with the item (e.g., "product", "service").
- **tags:** Collection of tags associated with the item, containing between 1 and 5 unique keywords.
- **related_items:** Collection of item_ids for referencing related items, which can contain duplicates.
- **description:** Contains either a short or long description of the item. The system must distinguish whether the description is short or long and store the associated description text accordingly.

Develop the Cassandra CQL code to create a table (column family) that meets these requirements.

QUESTION 2 [18 MARKS]

Scenario: Monitoring Industrial Machine Performance

System: Industrial Machine Monitoring System

Data: Sensor readings from various sensors attached to industrial machines on a factory floor.

Challenge: Each machine generates a variety of data points crucial for monitoring performance and identifying potential issues. These include:

- Machine ID (unique identifier for each machine)
- Vibration levels (measured at different points on the machine)
- Temperature readings (of various machine components)
- Power consumption

- Operational status (running, idle, error)
- Timestamp of the reading

Table 1: Test Data

Machine_ID	Data Recorded	
machine_1	vibration: • motor: 0.5, • gearbox: 0.2 temperature: • engine: 85.0, • coolant: 70.0 power_consumption: 1200.0, status: running timestamp: 2024-05-20 00:00:00	vibration: • motor: 0.4, • gearbox: 0.3 temperature: • engine: 87.0, • coolant: 72.0 power_consumption: 1150.0, status: running timestamp: 2024-05-20 01:00:00
machine_2	vibration: • motor: 0.7, • bearing_01: 0.1 temperature: • engine: 90.0, • coolant: 65.0 power_consumption: 1500.0, status: running timestamp: 2024-05-20 00:00:00	

2.1.Explain how you will model the system using Cassandra

[6 Marks]

2.2.Write the Cassandra code to implement your solution in 1.1 and insert the data in Table 1.

[5 marks]

2.3.Write the query to do the following:

2.3.1. Get the Latest Reading for a Machine_1

[2 Marks]

2.3.2. Get the Average Power Consumption for Each Machine

[3 Marks]

2.3.3. Find Machines with High Motor Vibration Levels

[2 Marks]

QUESTION 3 [25 MARKS]

Scenario

You are a software developer for a company that sells consumer goods. The company uses a relational database management system (RDBMS) to store its data. The company's data is currently stored in an SQL database. The RDBMS schema is shown in Figure 1. However, the company's data is growing rapidly, and the RDBMS is struggling to keep up. The company has decided to migrate its data to a NoSQL database management system (DBMS).

You have been tasked with migrating the company's data to a Neo4j graph database. Neo4j is a popular NoSQL DBMS that is well-suited for storing and querying interconnected data.

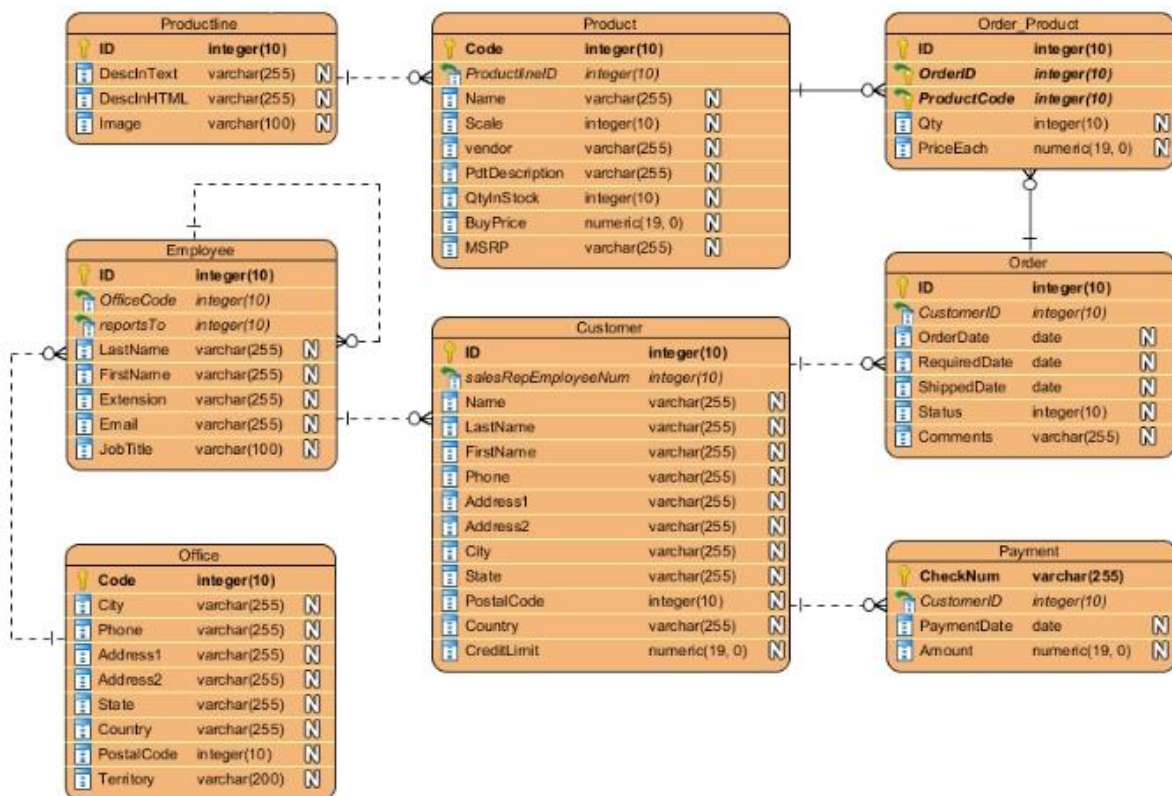


Figure 1: Database Schema

- 3.1. Model the System using Neo4J using the test data in Appendix A. Explain the choices you make in modelling the system. Submit the code used to create the graph, choices made and image of the graph.

[15 Marks]

- 3.2. Write the Neo4j queries to achieve the following (Submit your code and screen capture of the output):

3.2.1. Find all products and their product lines

[2 Marks]

3.2.2. Get the total number of employees in each office

[2 Marks]

3.2.3. Find all products included in Order 001. Display the Product Name and Product Code

[2 Marks]

3.2.4. Get the top 3 products by sales quantity

[4 Marks]

THE END

[Total: 50 MARKS]